Part C — Interview Ready (20%)

Q1 — What is Logistic Regression?

Logistic Regression is a classification algorithm — the name is misleading. It predicts the probability that an input belongs to a class using the sigmoid function, which squashes any real number into a value between 0 and 1. A threshold (usually 0.5) then converts that probability to a class label.

So even though it has 'regression' in the name, it's used for classification tasks — binary ones primarily.

Q2 — Train-Test Split and Scaling

```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

sc = StandardScaler()
X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)
```

Q3 — Confusion Matrix

A confusion matrix is a 2x2 table for binary classification:

	Predicted: Yes	**Predicted: No**
Actual: Yes	TP (True Positive)	FN (False Negative)
Actual: No	FP (False Positive)	TN (True Negative)

TP and TN are correct predictions. FP means the model said yes but was wrong. FN means the model missed an actual positive — often the more costly error in business contexts.